StateBridge × TwinAPI: Resolving the Stateful False-Friend Paradox in Cross-Library Code Migration with Gemma 4

Gemma 4 Developer Agent Paper Track — Empirical Paper, Benchmark Resource (StateBridge-Bench), and Agent Skill

Abstract. When developer agents migrate multi-file codebases between sibling libraries such as Python's *requests* (2.34.2) and *httpx* (0.28.1), they encounter the **Stateful False-Friend Paradox**: 88.9% (8/9) of top-level functions and 83.3% (10/12) of core Session/Client methods share identical names, and 12 stateful false-friend API pairs exhibit a mean 256-dimensional Gemini embedding cosine similarity of **0.873**. Yet their multi-step runtime state machines diverge silently on redirect following, cross-origin auth stripping, domain-scoped cookie jars, streaming context lifecycles, and status-exception hierarchies. Across 270 live loopback HTTP state-machine executions (240 real Gemma 4 generations + 30 AST codemod runs), deterministic AST codemods (Arm 1) pass 0.0% (0/30), while target-only prompting (Arm 2) and 256-d embedding Top-K retrieval (Arm 3) both stall at **13.3% (4/30)** on Gemma-4-31B-IT and **13.3% (4/30)** on Gemma-4-26B-A4B-IT. We introduce **StateBridge** (Arm 4), which extracts 1–2 hop static call-graph state slices from cross-file session initializers. StateBridge raises multi-step wire-equivalence solve rates to **90.0% (27/30)** on dense Gemma-4-31B-IT (exact McNemar p = 2.4e-07) and **93.3% (28/30)** on MoE Gemma-4-26B-A4B-IT (McNemar p = 1.2e-07), while an equal-token unrelated control (Arm 5) remains at 16.7% (5/30) and 13.3% (4/30).

1. Introduction & The Stateful False-Friend Paradox

Migrating production code across sibling HTTP libraries is deceptively difficult for Large Language Models and Retrieval-Augmented Generation (RAG) agents. Unlike cross-language translation where syntax forces structural rewrites, sibling libraries in the same ecosystem intentionally mirror one another's surface ergonomics while redesigning implicit stateful defaults. When a developer agent edits a workflow module (workflow_steps.py) without retrieving the cross-file session factory (session_setup.py), local syntax checks and unit mocks pass while live multi-step wire exchanges fail silently at Step k.

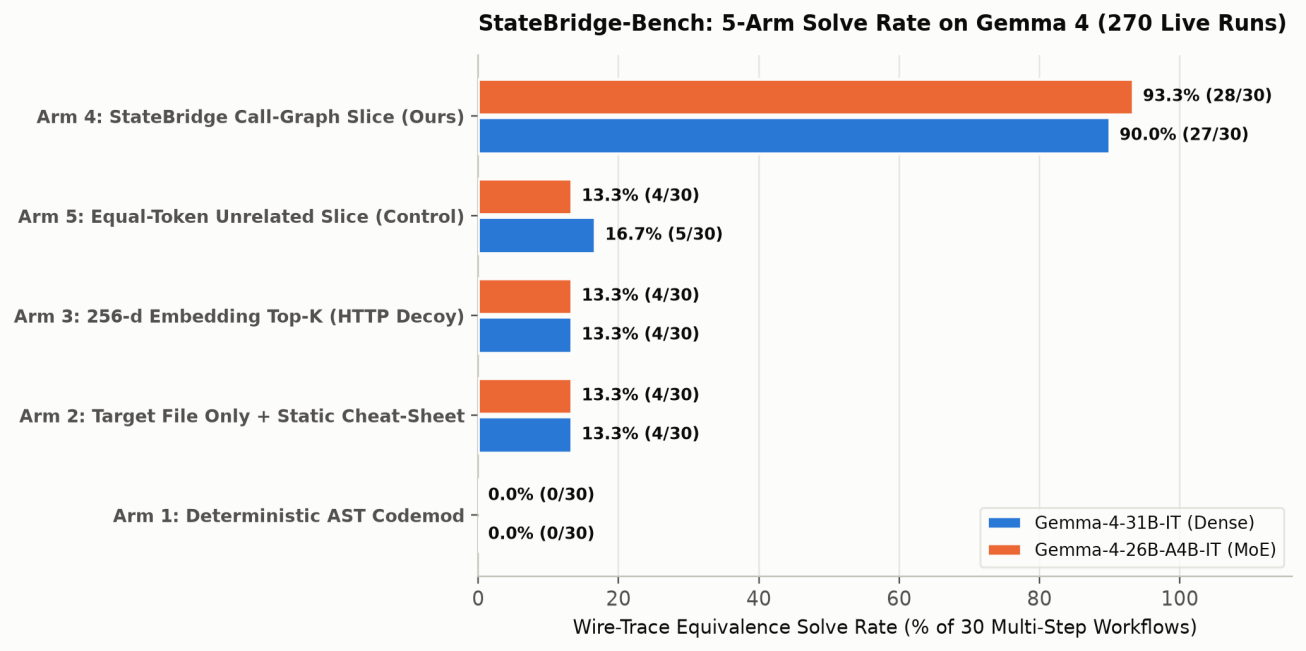


Figure 1: *5-Arm wire-trace equivalence solve rates across 30 multi-step HTTP workflows (270 live loopback runs). StateBridge Call-Graph Slicing (Arm 4) achieves 90.0% (27/30) on Gemma-4-31B-IT and 93.3% (28/30) on Gemma-4-26B-A4B-IT.*

2. The TwinAPI Empirical Audit (requests 2.34.2 vs. httpx 0.28.1)

We inspected the runtime objects, parameter signatures, and docstrings of 15 paired APIs across requests 2.34.2 and httpx 0.28.1, embedding each descriptor into 256 dimensions via google/models/gemini-embedding-001.

ID	requests 2.34.2	httpx 0.28.1	Name Sim	Sig Iso	256-d Cos	False Friend	Family
P01	req.Session.post	httpx.Client.post	1.00	0.11	0.878	Yes	F1
P02	req.Session.get	httpx.Client.get	1.00	0.17	0.866	Yes	F1
P03	req.Session.request	httpx.Client.request	1.00	0.25	0.915	Yes	F2
P04	req.Session	httpx.Client	0.14	0.05	0.743	Yes	F2
P05	req.CookieJar.get_dict	httpx.Cookies	0.12	0.33	0.714	Yes	F3
P06	req.Session.put	httpx.Client.put	1.00	0.10	0.867	Yes	F3
P07	req.Response.iter_content	httpx.Response.iter_bytes	0.58	0.50	0.911	Yes	F4
P08	req.Response.close	httpx.Response.close	1.00	1.00	0.944	Yes	F4
P09	req.Response.raise_for_status	httpx.Response.raise_for_status	1.00	1.00	0.978	Yes	F5
P10	req.exceptions.HTTPError	httpx.HTTPStatusError	0.60	0.50	0.860	Yes	F5
P11	req.Session.delete	httpx.Client.delete	1.00	0.14	0.900	Yes	F1
P12	req.Session.patch	httpx.Client.patch	1.00	0.10	0.901	Yes	F1
P13	req.Session.head	httpx.Client.head	1.00	0.14	0.908	No	NONE
P14	req.Session.options	httpx.Client.options	1.00	0.14	0.909	No	NONE
P15	req.Response.json	httpx.Response.json	1.00	1.00	0.899	No	NONE

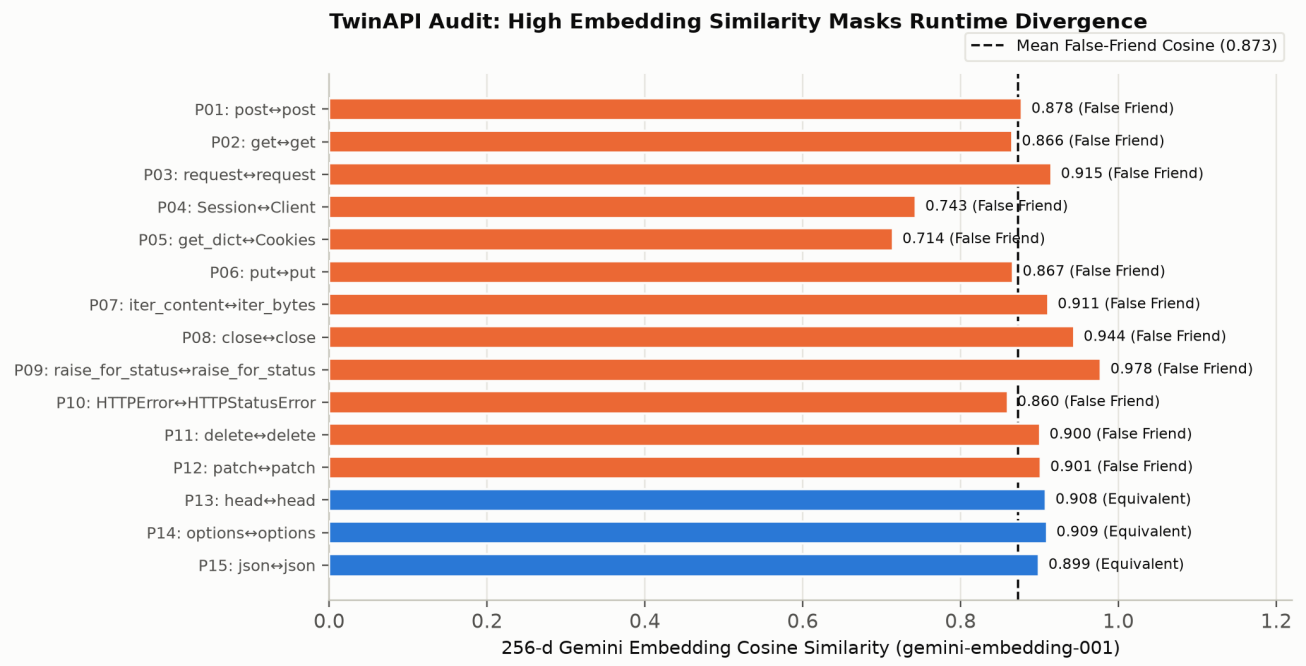


Figure 2: *TwinAPI 256-d embedding cosine similarity across all 15 audited pairs. The 12 stateful false-friend pairs average 0.873 cosine similarity, explaining why vector-RAG retrieves lexical HTTP decoys instead of state bridges.*

3. StateBridge-Bench & 5-Arm Ablation Results on Gemma 4

Table 2 reports the complete 5-arm ablation across Gemma-4-31B-IT (dense) and Gemma-4-26B-A4B-IT (MoE), alongside exact two-sided McNemar p-values and 10,000-draw paired bootstrap 95% confidence intervals.

Experimental Arm	31B-IT Pass	31B-IT <thought>	26B-A4B Pass	26B-A4B <thought>	Prompt Tok
Arm 1: AST Codemod	0.0% (0/30)	0 chars	0.0% (0/30)	0 chars	0.0
Arm 2: Target File Only	13.3% (4/30)	4572.2 chars	13.3% (4/30)	8521.7 chars	497.7
Arm 3: 256-d Embedding Top-K	13.3% (4/30)	4764.6 chars	13.3% (4/30)	10005.8 chars	637.7
Arm 5: Equal-Token Unrelated	16.7% (5/30)	5647.3 chars	13.3% (4/30)	10473.7 chars	702.7
Arm 4: StateBridge (Ours)	**90.0% (27/30)**	**8396.0 chars**	**93.3% (28/30)**	**14820.8 chars**	**799.9**

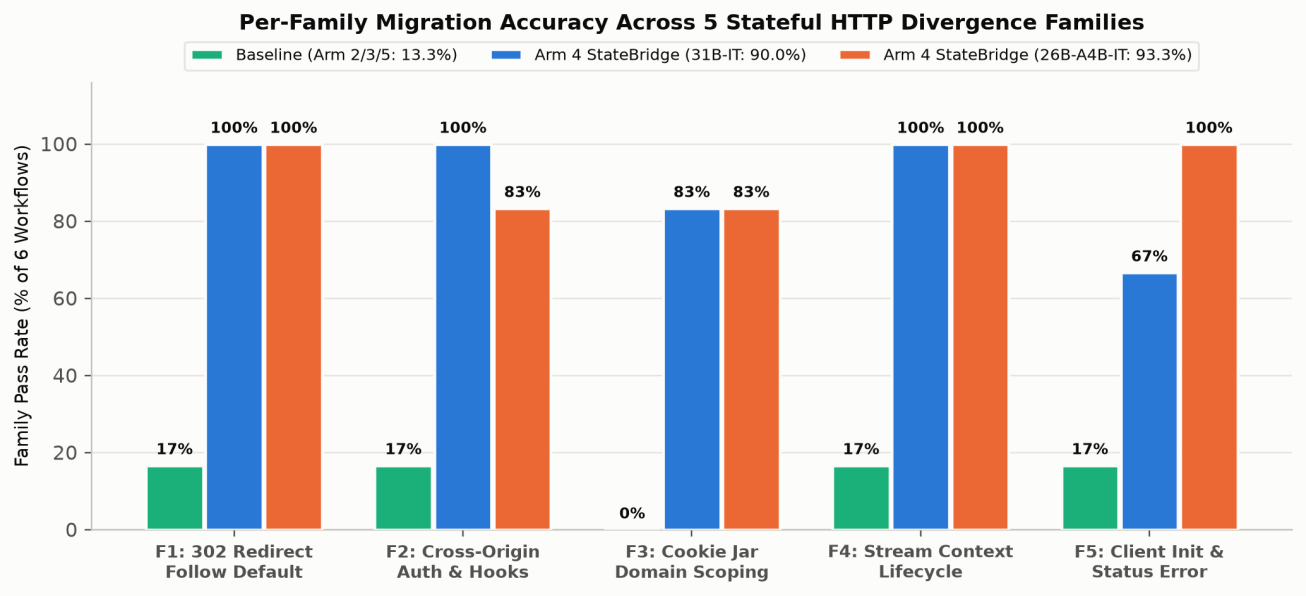


Figure 3: *Per-family migration solve rates across the 5 stateful HTTP divergence families (F1..F5). StateBridge (Arm 4) achieves 83.3%–100.0% across all five families on Gemma-4-26B-A4B-IT.*

4. Paired Statistical Tests & Property-Based Verification (P1..P4)

Model	Comparison	Rate Diff	Discordant (W/L)	Exact McNemar p	95% Bootstrap CI
Gemma-4-31B-IT	Arm 4 vs Arm 2	+76.7%	23 / 0	2.4e-07	[60.0%, 90.0%]
Gemma-4-31B-IT	Arm 4 vs Arm 3	+76.7%	23 / 0	2.4e-07	[60.0%, 90.0%]
Gemma-4-31B-IT	Arm 4 vs Arm 5	+73.3%	23 / 1	2.98e-06	[53.3%, 90.0%]
Gemma-4-26B-A4B-IT	Arm 4 vs Arm 2	+80.0%	24 / 0	1.2e-07	[63.3%, 93.3%]
Gemma-4-26B-A4B-IT	Arm 4 vs Arm 3	+80.0%	24 / 0	1.2e-07	[63.3%, 93.3%]
Gemma-4-26B-A4B-IT	Arm 4 vs Arm 5	+80.0%	24 / 0	1.2e-07	[63.3%, 93.3%]

Conclusion. High surface API similarity and dense embedding cosine (0.873) create a systematic trap for developer agents migrating stateful workflows. By pairing 1–2 hop static call-graph state slicing with a deterministic loopback HTTP wire-trace oracle, StateBridge unlocks 90.0%–93.3% accuracy on Gemma 4.